Huawei Access Node WHI-ACC-81 Whitefield East POP Routing Protocol Flap Resolution SOP

Node: WHI-ACC-81 | **Vendor:** Huawei | **Severity:** Critical | **Alarm Code:** ALARM_3174

Description

This Standard Operating Procedure (SOP) outlines the steps to be taken in the event of a critical routing protocol flap failure on Huawei Access Node WHI-ACC-81 located at the Whitefield East POP. The failure domain in question involves a BGP route churn, OSPF adjacency loss, and ISIS LSP flooding, which is indicative of a critical issue that necessitates immediate attention. The root cause of this failure domain is often attributed to misconfiguration, hardware failure, or software anomalies within the routing protocol stack. As a result, it is essential to employ a systematic approach to diagnose and resolve the issue, ensuring that all necessary precautions are taken to prevent further network disruption. The Huawei Access Node WHI-ACC-81 is a critical component of the Whitefield East POP's infrastructure, and any failure in this node can have a significant impact on network performance and availability. Therefore, it is essential to respond to this failure domain with the utmost urgency and diligence.

Pre-requisite Checks

- Check: Verify redundant link status to ensure no single point of failure
- Check: Confirm that remote hands are dispatched to the Whitefield East POP for assistance
- Check: Ensure that all necessary diagnostic tools and equipment are available on site

Potential Root Causes

Cause	Probability
Misconfigured BGP peers	High
Faulty OSPF interface configuration	Medium
Hardware failure in the routing protocol stack	Low

Diagnostic Commands & Expected Outputs

```
Command: display bgp peer, Expected Output: BGP Peer Information, Description: This command displays information about BGP peers, including their status and any errors that may have occurred.

Command: display ospf interface, Expected Output: OSPF Interface Information, Description: This command displays information about OSPF interfaces, including their status and any errors that may have occurred.

Command: display isis lsp, Expected Output: ISIS LSP Information, Description: This command displays information about ISIS LSPs, including their status and any errors that may have occurred.
```

Resolution Steps

Step 1: Step: Verify BGP peer configuration, Command: display bgp peer, Description: Verify that BGP peer configuration is correct and consistent across all peers.

Step 2: Step: Reset OSPF interface, Command: reset ospf interface, Description: Reset OSPF interface to ensure that it is functioning correctly.

Step 3: Step: Replace faulty hardware, Command: replace hardware, Description: Replace faulty hardware in the routing protocol stack to ensure that it is functioning correctly.

Post-Check Verification

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Command: display bgp peer, Expected Output: BGP Peer Information, Description: Verify that BGP peer configuration is correct and consistent across all peers.

Command: display ospf interface, Expected Output: OSPF Interface Information, Description: Verify that OSPF interface is functioning correctly.

Command: display isis lsp, Expected Output: ISIS LSP Information, Description: Verify that ISIS LSPs are functioning correctly.
```

Roll-back Procedures

- Step: Revert to previous configuration, Command: undo bgp peer, Description: Revert to previous BGP peer configuration in the event of a failure.

- Step: Revert to previous OSPF interface configuration, Command: undo ospf interface, Description: Revert to previous OSPF interface configuration in the event of a failure.

- Step: Revert to previous hardware configuration, Command: undo replace hardware, Description: Revert to previous hardware configuration in the event of a failure.